

# BLUEPRINTIQ TECHNICAL MEMORANDUM

UNCERTAINTY-AWARE BLUEPRINT-TO-BOQ INTELLIGENCE | DETAILED ANALYSIS REPORT

**STATUTORY ENGINEERING NOTICE:** BlueprintIQ provides automated preliminary computer-vision geometry extraction and material estimations. This document does not replace a licensed professional engineer, architect, quantity surveyor, or municipal authority approval. All quantities must be field-verified before procurement.

## 1. PROJECT & DRAWING METADATA

|                 |          |                     |              |
|-----------------|----------|---------------------|--------------|
| Project Name:   | villa 3  | Building Type:      | Commercial   |
| Floors:         | 2        | Unit System:        | METRIC       |
| Target Area:    | 180.0 m² | Soil Type:          | Not Provided |
| Drawing Scale:  | Unknown  | Concrete Grade:     | M20          |
| Wall Thickness: | 230 mm   | Overall Confidence: | MEDIUM       |

## 2. CONSOLIDATED MATERIAL ESTIMATES (UNCERTAINTY-AWARE)

| MATERIAL                               | ESTIMATED    | RANGE          | CONF.  | CALCULATION BASIS                                                     |
|----------------------------------------|--------------|----------------|--------|-----------------------------------------------------------------------|
| Cement (OPC/PPC 50kg Bags)             | 645.0 bags   | 594 - 710      | MEDIUM | Concrete (557.4 bags) + Masonry (36.3 bags) + Plaster (51.5 bags)     |
| River Sand / M-Sand                    | 45.8 m³      | 41 - 51        | MEDIUM | Concrete (29.0m³) + Masonry (7.6m³) + Plaster (9.2m³)                 |
| Coarse Aggregate (20mm & 10mm Graded)  | 58.7 m³      | 53 - 65        | MEDIUM | RCC Concrete (58.1m³) + PCC Bed (0.7m³)                               |
| Modular Clay / Fly Ash Bricks          | 10,567.0 nos | 9,721 - 11,624 | HIGH   | 20.13 m³ net masonry x 500 bricks/m³ + 5% breakage allowance          |
| Reinforcement Steel (Fe500 TMT Rebars) | 5,172.0 kg   | 4,396 - 6,464  | MEDIUM | Slabs (3391.0 kg at 0.8% vol) + Beams/Columns (1780.0 kg at 1.5% vol) |
| Vitrified Floor Tiles (600x600mm)      | 159.4 m²     | 151 - 175      | MEDIUM | Net floor area 0m² + 8% cutting and breakage allowance                |
| Emulsion Paint & Primer (2 Coats)      | 90.0 litres  | 81 - 104       | MEDIUM | Total plaster & ceiling area 900.0m² / coverage 10 m²/Litre           |

## 3. DETAILED BILL OF QUANTITIES (CIVIL & FINISHES)

| CODE   | DESCRIPTION                                                 | QTY    | UNIT | RANGE         | CONF.  |
|--------|-------------------------------------------------------------|--------|------|---------------|--------|
| CIV-01 | Earthwork excavation in foundation trenches                 | 11.14  | m³   | 7.8 - 16.2    | LOW    |
| CIV-02 | Plain Cement Concrete (PCC 1:4:8) in foundation bed (100mm) | 0.81   | m³   | 0.7 - 0.9     | MEDIUM |
| CIV-03 | Reinforced Cement Concrete (M20) for Slabs, Beams, Columns  | 69.12  | m³   | 63.6 - 77.4   | MEDIUM |
| CIV-04 | Brick Masonry in Cement Mortar (1:6) with Modular Bricks    | 20.13  | m³   | 18.1 - 22.1   | HIGH   |
| CIV-05 | Cement Plastering (12mm Internal 1:6 + 18mm External 1:4)   | 540.00 | m²   | 502.2 - 594.0 | MEDIUM |
| CIV-06 | Flooring with Vitrified Tiles including 8% wastage          | 159.41 | m²   | 151.4 - 172.2 | MEDIUM |

## 4. BLUEPRINT ANOMALIES & CODE COMPLIANCE FINDINGS

| ID        | SEV. | FINDING & EVIDENCE                                                                                                                                                                                              | RECOMMENDED ACTION                                                                                                                       |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| ISSUE-001 | HIGH | <b>Drawing Scale Not Confirmed or Uncalibrated</b><br>Drawing scale specified as 'Unknown' without dimension calibration points.<br><i>Standard: IS 1200 / ISO 128 Architectural Drawing Presentation Rules</i> | Perform scale calibration against a known reference dimension on the drawing before procurement.<br><b>Status: Verification Required</b> |

|           |      |                                                                                                                                                   |                                                                                                           |
|-----------|------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| ISSUE-002 | HIGH | <b>Sub-Standard Door Opening Width (Door (91%))</b><br>Detected door span: 0.60 m<br><i>Standard: IBC 2021 Section 1010.1.1 / NBC 2016 Part 3</i> | Verify rough opening (RO) size to ensure ADA / egress compliance.<br><b>Status: Verification Required</b> |
| ISSUE-003 | HIGH | <b>Sub-Standard Door Opening Width (Door (86%))</b><br>Detected door span: 0.60 m<br><i>Standard: IBC 2021 Section 1010.1.1 / NBC 2016 Part 3</i> | Verify rough opening (RO) size to ensure ADA / egress compliance.<br><b>Status: Verification Required</b> |
| ISSUE-004 | HIGH | <b>Sub-Standard Door Opening Width (Door (78%))</b><br>Detected door span: 0.60 m<br><i>Standard: IBC 2021 Section 1010.1.1 / NBC 2016 Part 3</i> | Verify rough opening (RO) size to ensure ADA / egress compliance.<br><b>Status: Verification Required</b> |
| ISSUE-005 | HIGH | <b>Sub-Standard Door Opening Width (Door (73%))</b><br>Detected door span: 0.60 m<br><i>Standard: IBC 2021 Section 1010.1.1 / NBC 2016 Part 3</i> | Verify rough opening (RO) size to ensure ADA / egress compliance.<br><b>Status: Verification Required</b> |
| ISSUE-006 | HIGH | <b>Sub-Standard Door Opening Width (Door (63%))</b><br>Detected door span: 0.60 m<br><i>Standard: IBC 2021 Section 1010.1.1 / NBC 2016 Part 3</i> | Verify rough opening (RO) size to ensure ADA / egress compliance.<br><b>Status: Verification Required</b> |
| ISSUE-007 | HIGH | <b>Sub-Standard Door Opening Width (Door (58%))</b><br>Detected door span: 0.60 m<br><i>Standard: IBC 2021 Section 1010.1.1 / NBC 2016 Part 3</i> | Verify rough opening (RO) size to ensure ADA / egress compliance.<br><b>Status: Verification Required</b> |
| ISSUE-008 | HIGH | <b>Sub-Standard Door Opening Width (Door (29%))</b><br>Detected door span: 0.60 m<br><i>Standard: IBC 2021 Section 1010.1.1 / NBC 2016 Part 3</i> | Verify rough opening (RO) size to ensure ADA / egress compliance.<br><b>Status: Verification Required</b> |
| ISSUE-009 | HIGH | <b>Sub-Standard Door Opening Width (Door (21%))</b><br>Detected door span: 0.60 m<br><i>Standard: IBC 2021 Section 1010.1.1 / NBC 2016 Part 3</i> | Verify rough opening (RO) size to ensure ADA / egress compliance.<br><b>Status: Verification Required</b> |

## 5. LOCAL AI (GEMMA 2B) SYNTHESIS & REASONING

**Executive Synthesis:** Preliminary architectural evaluation performed for Commercial project. Detected 9 potential dimensional or compliance considerations (9 require urgent review). Soil parameter was 'Not Provided', requiring geotechnical confirmation before finalizing foundation estimates.

**Uncertainty Assessment:** Overall analysis confidence is rated as MEDIUM. Confidence is constrained by 2D plan annotations and missing structural section schedules.

## 6. MISSING PARAMETERS & ENGINEERING ASSUMPTIONS

### Missing Input Parameters:

- **Soil Type:** Not Provided — *Foundation trench excavation, footing depth, and sub-base quantities cannot be verified without soil bearing capacity (SBC).*
- **Scale Calibration:** Uncalibrated — *Drawing scale relies on nominal text ratio ('1:100') rather than two-point physical dimension calibration. Coordinate errors may magnify calculated areas by 5-10%.*

### Applied Engineering Assumptions:

- Excavation depth assumed at nominal 1.2m for medium dense soil; footing sizes subject to soil report.
- Graphic scale ratio Unknown assumed uniformly proportional across X and Y axes.
- Reinforcement steel estimated via standard IS 456 thumb rules (0.8% for slabs, 1.5% for framed elements).